INCH-POUND
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DETAIL SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, DELAY M4A1, M12A1, M45A1, and M49A1 REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M4A1, M12A1, M45A1, and M49A1 Delay Cartridge Actuated Initiator Remanufactured Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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STANDARDS

MILITARY

MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

(Unless otherwise indicated, copies of military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

8595817	Initiator, Cartridge Actuated, Delay, Metal Parts Assembly
8796826	Box, Fiberboard (Inner) Packing, Ammunition, for Initiator, Cartridge Actuated, Delay, M4A1, M6A1, M12A1, M26, M33, M44, M45A1, or M49A1
8796827	Box, Packing, Ammunition, for Initiator, Cartridge Actuated, Delay, M4A1, M6A1, M12A1, M26, M33, M44, M45A1, or M49A1
9216824	Box, Fiberboard (Outer) Alternative, Packing, Ammunition, for Initiator, Cartridge Actuated, Delay, M4A1, M6A1, M12A1, M26, M33, M44, M45A1, or M49A1

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL

ASQC Z1.4	Sampling Procedures and Tables for Inspection by Attributes
ASQC A8402	Quality Management and Quality Assurance—Vocabulary

(Application for copies should be addressed to the American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. The remanufactured metal parts assembly shall contain a new safety pin, initiator spring, O-rings, and seal.

3.2 Hydrostatic pressure. The initiator cap, chamber, and plug, simulating the firing pin housing (without a firing pin pin hole but with an O-ring), shall withstand an internal hydrostatic pressure of 5500 to 6500 pounds per square inch (psi) for 15 seconds minimum, without deformation or mechanical failure (see 6.4.1).

3.3 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component, regardless of configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied (see 6.4.2).

3.4 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than 5 degrees in either direction (see 4.5.5).

3.5 Firing pin release. The firing pin of the assembled initiator, less safety pin, cartridge retainer and chamber, shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin 3/4 inches, maximum.

3.6 Initiator pin security. The initiator pin of the assembled initiator, less cartridge retainer and chamber, shall not withdraw beyond the safety pin when subjected to an axial load of 200 to 220 pounds. Application of the load shall cause no deformation of the safety pin.

3.6.1 Initiator pin. Check safety pin groove length for 0.118 ± 0.010 inch dimension (100 percent inspection.)

3.7 Metal parts assembly. Parts shall be combined to produce an assembly that conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.7.1 Integrated plant products. When remanufacturing and loading of the assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

3.7.2 Threaded component parts. Class 2A or 2B thread gage shall be used on component part inspection versus class 3A or 3B thread gage required on remanufactured hardware for initiators.

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3.8 Workmanship. Assemblies shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Quality assurance terms and definitions. Reference shall be made to ASQC A8402 to define the quality assurance terms used.

4.1.1 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient (68 to 77°F)
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with ASQC Z1.4.

4.3.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 assemblies maximum, provided the homogeneity requirement of inspection lots have been retained.

4.3.2 Examination. On reprocessed parts, one hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects, using the sampling plan and associated criteria specified in the contract (see 6.2 and 6.3). Four power (4x) magnification may be used for inspection of such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

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4.3.2.1 Examination of reclaimed parts. Examination of reusable parts that require no reprocessing shall be limited to inspecting only those defects requiring visual inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.3.2.2 Pin, Initiator, Drawing 8595403, covering a detail of Drawing 8595817.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Surface finish on body and body groove improper	Visual
104	Burr	Visual
105	Foreign matter	Visual
Minor		
201	Surface finish improper, except as otherwise classified	Visual

4.3.2.3 Cap, Drawing 8595404, covering a detail of Drawing 8595817.

Critical		
1	Metal defective	Visual
Major		
101	Pitch diameter of large thread, max	Gage
102	Minor diameter of large thread, max	Gage
103	Pitch diameter of small thread, max	Gage
104	Minor diameter of small thread, max	Gage
105	Thread damaged	Visual
106	Metal defective, except as otherwise classified	Visual
107	Protective finish inadequate or defective	Visual
108	Burr	Visual
109	Foreign matter	Visual
Minor		
None defined		

4.3.2.4 Housing, Firing Pin, Drawing 8595408, covering a detail of Drawing 8595817.

Critical		
1	Metal defective	Visual

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Major		
101	Pitch diameter of small thread, min	Gage
102	Major diameter of small thread, min	Gage
103	Pitch diameter of large thread, min	Gage
104	Major diameter of large thread, min	Gage
105	Thread damaged	Visual
106	Protective finish inadequate or defective	Visual
107	Surface finish in cavity improper	Visual
108	Burr or sharp corner	Visual
109	Foreign matter	Visual
Minor		
201	Surface finish improper, except as otherwise classified	Visual

4.3.2.5 Pin, Firing, Drawing 8595407, covering a detail of Drawing 8595817.

Critical
None defined

Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Surface finish improper	Visual
104	Burr or sharp edge	Visual
105	Foreign matter	Visual

Minor
None defined

4.3.2.6 Chamber, Initiator, Drawing 8593314, covering a detail of Drawing 8595817.

Critical		
1	Metal defective	Visual
Major		
101	Pitch diameter of large thread, min	Gage
102	Major diameter of large thread, min	Gage
103	Pitch diameter of small thread, min	Gage
104	Major diameter of small thread, min	Gage
105	Thread damaged	Visual
106	Protective finish inadequate or defective	Visual
107	Surface finish, end of gas port, improper	Visual
108	Burr	Visual
109	Foreign matter	Visual

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Minor		
201	Surface finish improper, except as otherwise classified	Visual

4.3.2.7 Retainer, Cartridge, Drawing 8593313, covering a detail of Drawing 8595817.

Critical		
1	Metering slots in base missing	Visual
Major		
101	Pitch diameter of thread, max	Gage
102	Minor diameter of thread, max	Gage
103	Elongated slots missing	Visual
104	Thread damaged	Visual
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Burr	Visual
108	Foreign matter	Visual
Minor		
None defined		

4.3.2.8 Assembly, Drawing 8595817, less all drawings except 8595408. 8595407, 8595403, 8595406, MS19060 and MS28775.

Critical		
1	Protrusion of firing pin	Gage
2	Steel ball missing	Gage
3	O-ring missing	Visual
Major		
None defined		
Minor		
201	Lubricant missing from O-ring	Visual

4.3.2.9 Assembly, Drawing 8595817.

Critical		
None defined		
Major		
101	Initiator pin not free to rotate	Manual
102	Safety pin loose or deformed	Manual/Visual
103	Any part missing	Visual
104	Improper assembly	Visual

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Minor

201	Protective cap missing	Visual
202	Protective finish inadequate or defective	Visual

4.3.2.10 Box, Fiberboard (Inner) Packing, Ammunition, Drawing 8796826.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.3.2.11 Box, Packing, Ammunition, Drawing 8796827.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.3.2.12 Box, Fiberboard (Outer) Alternative, Packing, Ammunition, Drawing 9216824.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual\Visual
202	Hardware missing, loose, broken, or improperly engaged	Visual
203	Strapping missing, loose, broken, or improperly engaged	Visual

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204	Contents of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect, or illegible	Visual

4.4 Inspection testing. The following tests shall be performed in accordance with 4.5.

4.4.1 Hydrostatic pressure. Each chamber and cap shall be subjected to the hydrostatic pressure test. Failure of any subassembly to comply with the requirement of 3.2 shall be cause for rejection of the subassembly.

4.4.2 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.4.3 Firing pin release. The initiator pin of each subassembly shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.5 shall be cause for rejection of the subassembly.

4.4.4 Initiator pin security. The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any subassembly to comply with the requirement of 3.6 shall be cause for rejection of the subassembly.

4.4.5 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism specified in 4.5.5. Failure of any unit in the inspection sample to comply with the requirement of 3.4 shall be cause for rejection of the assembly.

4.5 Test methods and procedures.

4.5.1 Hydrostatic pressure. The subassembly shall be mounted in a device using the fixture for properly sealing all openings and subjected to hydrostatic pressure to determine compliance with 3.2.

4.5.2 Firing pin release. The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

4.5.3 Initiator pin security. The subassembly shall be secured in the fixture with the initiator pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in each position.

4.5.4 Protective finish. The procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.5.5 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end, and at the same height as the compass, in the north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per

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minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.6 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).

c. Any special inspection conditions (see 4.2).

d. Production test activity and lot size (see 4.3.1.2).

e. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8796826 and 8796827. Level B shall be in accordance with Drawings 8796826 and 9216824.

f. Air Force packaging shall be in accordance with SPI 00-845-1058.

6.3 Inspection sampling plan and acceptance/rejection criteria. The sampling plan and acceptance/rejection criteria of tables I and II have been used successfully in procuring, testing, and examining the remanufactured Initiator, Cartridge Actuated, Delay, M4A1, M12A1, M45A1, and M49A1 Metal Parts Assembly. For an individual defect (single characteristic), the criteria in table I have been used.

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For total defects within a major or minor group, the criteria in table II have been used. The contracting activity is cautioned that any deviation from the values in table I and II should be warranted and verified statistically considering the criticality of the remanufactured metal parts assembly's function.

TABLE I. Sampling plan.

Lot size	Sample size	Major Defects		Minor Defects	
		Accept	Reject	Accept	Reject
1 to 300	100% Examination	—	—	—	—
301 to 1300	110	1	2	2	3
1301 to 3200	150	2	3	3	4
3201 to 8000	225	3	4	4	5
8001 to 22,000	300	4	5	5	6

TABLE II. Total allowable defects per sample size.

Sample size for major defects	110	150	225	300
Number of defects in class				
2 to 3	1	2	3	4
4 to 7	2	3	4	5
8 to 19	3	4	5	7
20 to 43	4	5	8	10
Over 43	6	8	11	14
Sample size for minor defects	110	150	225	300
Number of defects in class				
2 to 3	2	3	4	5
4 to 7	3	4	5	7
8 to 16	4	5	8	10
17 to 48	6	8	11	14
Over 48	8	11	17	20

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6.4 Definitions.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.4.2 Exterior components. Remanufactured exterior components are: cap, initiator pin, and chamber.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Preparing activity
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